IC103 Lecture 3: Key Takeaways

Theory of Production

1. Production and Inputs

- **Firms combine** inputs to produce **output.**
- A production function shows this relationship.
- General form: $Q = f(X_1, X_2, ..., X_n)$.
- A simpler example is $Q = f(L, K)$, where L = labour and K = capital.



Output depends on the quantity and combination of inputs used.

2. Fixed and Variable Inputs

1 Fixed inputs

- Cannot be changed easily in the short run.
- Examples: buildings, machinery, factory space.



2 Variable inputs

- Can be changed in the short run.
- Examples: labour, raw materials, electricity.



Whether an input is fixed or variable depends on the time period considered.

3. Short Run and Long Run

- **Short run:** at least one input is fixed.
- **Long run:** all inputs can be varied.
- **In the** long run, the firm can change its scale of production.

Short run



At least one input is fixed.

Long run



All inputs can be varied (including scale).

Short run does not mean a short calendar period — it means some inputs are fixed.

4. Diminishing Marginal Returns

- When more units of a variable input are added to fixed inputs, each extra unit eventually adds less extra output than the previous one.
- Total output may still rise, but at a slower rate.
- This is a short-run concept.

Example: a coffee shop

- One espresso machine is fixed (in the short run).
- Baristas are variable.
- Adding more baristas initially increases output a lot, but each additional barista eventually adds less extra coffee.



Diminishing returns means extra output falls — not necessarily total output.

5. TP, AP and MP

- **Total Product (TP):** total output produced.
- **Average Product (AP):** output per unit of variable input.
- **Marginal Product (MP):** extra output from one more unit of variable input.

Average product (AP)

$$AP = \frac{TP}{L}$$

Marginal product (MP)

$$MP = \frac{\Delta TP}{\Delta L}$$

MP tells us how much extra output is produced by one more worker.

6. What Happens as Labour Increases?

Workers (L)	Total Product (TP)	Marginal Product (MP)
1	20	20
2	45	25
3	62	17
4	70	8
5	73	3

- At first, output rises quickly as more labour is added.
- Later, marginal product falls.
- This illustrates diminishing marginal returns.

As labour increases, MP eventually falls because fixed inputs become a constraint.

Quick Revision Checklist

- ☐ Define a production function.
- ☐ Explain fixed and variable inputs.
- ☐ Distinguish between the short run and the long run.
- ☐ Explain diminishing marginal returns.
- ☐ Define TP, AP and MP.